

Heat — Workout

ICSE Class 7 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Whole chapter: heat and temperature, scales and conversions, thermal expansion, and conduction, convection and radiation.

Instructions: Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

1. A temperature of 35°C on the Fahrenheit scale is:

- | | |
|----------|---------|
| A. 63°F | B. 67°F |
| C. 308°F | D. 95°F |

2. Absolute zero, the lowest possible temperature, is:

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|----------|-----------|
| A. 273 K | B. 0°C |
| C. 0 K | D. -100°C |

3. Gaps in railway lines, rollers under bridges and loops in pipelines all allow for:

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|---------------------|--------------------------------------|
| A. bending of light | B. thermal expansion and contraction |
| C. rusting | D. electric current |

4. A device that uses a bimetallic strip to switch a circuit on and off at a set temperature is a:

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|-----------------|---------------|
| A. galvanometer | B. periscope |
| C. barometer | D. thermostat |

5. In conduction, heat travels through a solid because:

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|---|---------------------|
| A. the particles move from the hot end to the cold end | B. radiation occurs |
| C. heat passes from particle to particle without the particles moving along | D. the solid melts |

6. Smoke rises up a chimney because of ____ currents of hot air.

- | | |
|---------------|---------------|
| A. convection | B. conduction |
| C. radiation | D. magnetic |

7. Unlike conduction and convection, radiation:

- | | |
|----------------------------|-----------------------------|
| A. needs a solid | B. needs no material medium |
| C. cannot pass through air | D. needs a liquid |

8. A shiny silver surface is a ____ absorber and a ____ emitter of heat radiation.

- | | |
|---------------|---------------|
| A. poor, poor | B. poor, good |
| C. good, poor | D. good, good |

9. A room heater warms the air of a room mainly by radiation and by setting up ____ currents.

- | | |
|-------------|---------------|
| A. sound | B. convection |
| C. magnetic | D. conduction |

10. A thermos flask keeps cold drinks cold as well as hot drinks hot because it:

- | | |
|---|--------------------------|
| A. reduces heat transfer in both directions | B. adds heat |
| C. produces cold | D. only keeps things hot |

Answer Key

1. D

2. C

3. B

4. D

5. C

6. A

7. B

8. A

9. B

10. A